



Least Common Multiple & Greatest Common Factor

Prime Atoms, Venn Strategy, and Efficiency

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Session Flow: From Cycles to Building Blocks

A 50-minute student-centered sequence based on Buckingham's PGC Framework.

Lesson Roadmap (50 min)

- **05 min – Phase 1:** Novelty & Curiosity (The Syncing Lights Puzzle)
- **10 min – Phase 2:** Expectation & Motivation (LCM Tables vs. GCF Limits)
- **20 min – Phase 3:** Effort & Scaffolding (Prime Trees, Venn Strategy for LCM/GCF)
- **15 min – Phase 4:** Recognition & Peer Audit (Large Number Challenge & Error Detection)

1. Novelty & Curiosity | The Syncing Lights Puzzle

Driving Question

*"Light A flashes every 4 seconds. Light B flashes every 6 seconds.
If both flash together at $t = 0$, at what exact second will they sync again?"*

Visualizing Cycles

Light A (4s): ● ——— ● ——— ● ———

Light B (6s): ● ———— ● ————

Predict without using any formulas!

Think-Pair-Share (3 min)

- Share your initial prediction with your partner.
- Sketch a timeline or skip-counting list in your notebook to verify.

2. Expectation & Motivation | Defining the LCM

Task (Pair Discovery - 4 min): List the multiples of 4 and 6 to find where they overlap.

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Multiples Table

- **Multiples of 4:**

4, 8, **12**, 16, 20, **24**, 28, ...

- **Multiples of 6:**

6, **12**, 18, **24**, 30, ...

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12, 24, 36, ...

Least Common Multiple (LCM):

$$\text{LCM}(4, 6) = 12$$

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The Challenge Question

"What if we need to find the LCM of 48 and 72? Is writing full tables efficient?"

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[Level: Basic – Guided Discovery | Individual → Pair]

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Example: Find the GCF of 12 and 18

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Step 1: List All Factors

- **Factors of 12:** 1, 2, 3, **6**, 12
- **Factors of 18:** 1, 2, 3, **6**, 9, 18

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Step 3: Pick the Greatest (GCF)

$$\text{GCF}(12, 18) = \mathbf{6}$$

"6 is the biggest building block both numbers share!"

Quick Check (Pair Work - 1 min): Why is 3 a common factor, but NOT the GCF?

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Yes!

Why We Need a Strategy

Testing numbers one-by-one is slow and prone to errors.

Greatest Common Factor (GCF): The largest integer that divides two or more numbers without a remainder.

2. Expectation & Motivation | Primes & Fundamental Theorem

Guided Discovery: Deconstruct 12 into smaller multiplicative parts until you can go no further.

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An integer greater than 1 whose **only** positive factors are 1 and itself.

Primes: 2, 3, 5, 7, 11, 13, 17, ...

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Fundamental Theorem of Arithmetic

Every integer greater than 1 is either a **prime** or can be represented as a **unique product of prime numbers**.

Primes are the "atoms" of integers. Everything else is a compound!

3. Effort & Scaffolding | Level 3: Venn Diagram Strategy for LCM & GCF

[Level: Hard – Pair / Small Group Work | 8 min]

Goal: Use a Venn Diagram of Prime Factors to find both GCF and LCM for 18 and 24.

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The Rules

- **Intersection ($\cap$):** Shared prime factors $\rightarrow$ **GCF**
- **Union ($\cup$):** ALL prime factors $\rightarrow$ **LCM**

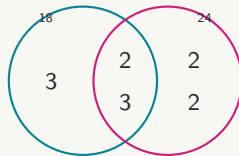
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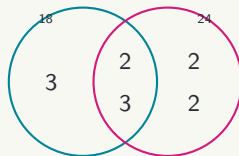
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- $\text{GCF}(18, 24) = 2 \times 3 = \mathbf{6}$
- $\text{LCM}(18, 24) = 3 \times (2 \times 3) \times 2 \times 2 = \mathbf{72}$

4. Recognition & Feedback | Peer Audit with Large Numbers

Challenge Task (8 min): Working individually, calculate both the **GCF** and **LCM** of the following large numbers using a Venn Diagram:

$$a = 120 \quad \text{and} \quad b = 180$$

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Challenge Task (8 min): Working individually, calculate both the **GCF** and **LCM** of the following large numbers using a Venn Diagram:

$$a = 120 \quad \text{and} \quad b = 180$$

Instructions for Peer Audit

Swap papers with your partner. Your partner must:

1. Verify if the prime factor trees for 120 ($2^3 \times 3 \times 5$) and 180 ($2^2 \times 3^2 \times 5$) are correct.
2. **Identify errors:** Point out if prime factors were placed in the wrong region of the Venn diagram or if GCF ($\cap$) vs. LCM ($\cup$) were confused.
3. Write a brief correction note and sign the page before submitting to the teacher.

Session Synthesis

Concept	Mathematical Meaning
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Venn Strategy	

Concept	Mathematical Meaning
LCM	The smallest shared multiple between numbers (Union $\cup$).
GCF	The largest shared factor between numbers (Intersection $\cap$).
Fundamental Theorem	Every integer > 1 decomposes uniquely into prime factors.
Venn Strategy	$\text{GCF} = \text{Product of Intersection}$ $\text{LCM} = \text{Product of Union}$.

Next Class: Adding and Subtracting Fractions with LCM!

50 Minutes of Focused Individual Work

Task Objectives & Instructions

- **Individual Focus:** Work independently on your assigned task/homework.
- **Self-Paced Progression:** Move sequentially through the problem sets or task guidelines.
- **Resource Management:** Consult your reference notes, class slides, or guidelines before asking questions.

Expectations

- Silent and continuous effort.
- Complete all required steps in your notebook/file.

Teacher Support

- Raise your hand for targeted 1-on-1 feedback.
- Check progress against the success criteria.